

Incorporating Fossils

Jeremy M. Beaulieu

As of version 2.1.14, we have added the ability for HiSSE, MuHiSSE, and MiSSE to handle fossils as well as stratigraphic ranges. The fossilized birth-death model (FBD) follows that of Stadler (2010), and includes a new parameter, ψ , that estimates the rate of fossils preservation in a birth-death model (see Beaulieu and O’Meara 2023). We have also extended the canonical FBD model to deal with fossil sets that lack sampled ancestors (fossilized birth-death of terminal fossils only, FBDT), the details of which will be included in a forthcoming paper (Beaulieu and O’Meara, Submitted). Below I will demonstrate how to set up a MiSSE model when the tree and data contain fossils. I will also demonstrate a simple test to show that both the HiSSE and MuHiSSE likelihoods are correct.

Before getting started, there are two important terms that need to be clearly understood. Under the canonical FBD there are actually two types of fossils that are sampled. The first are referred to as m fossils, which represent extinct tips – that is, they represent distinct branches that went extinct before the present. The second are referred k fossils, which represent samples of lineages that eventually made it to the present. In most real world cases, both fossil types must be present for the FBD to work well, though having k fossils is not absolutely critical for *hisse* whereas m fossils are. In cases where k fossils are absent *hisse* automatically defaults to the FBDT model.

Simulating a practice data set under MiSSE

Be sure to load the *hisse* and *TreeSim* packages. I will first simulate a tree with fossil samples under a simple fossilized birth-death model. This will be used as input into MiSSE, which sets up and executes a completely trait-free version of a HiSSE model (i.e., hidden states only). The first step is to simulate a tree that contains the complete history of the trees – that is, includes both taxa that survived to the present and those that did not:

```
set.seed(42)
phy <- TreeSim::sim.bd.taxa(n = 100, numbsim = 1, lambda = 0.3, mu = 0.2)[[1]]
```

In order to sample both extinct tips and edges, consistent under the fossilized birth-death model, we have written customized functions that takes as input the full tree and a value for ψ and returns a sampled tree:

```
f <- GetFossils(phy, psi=0.05)
fbd.tree <- ProcessSimSample(phy, f)
```

The first function, *GetFossils()* samples edges and extinct tips and provides a detailed table about these samples. The second function, *ProcessSimSample()*, takes the detailed table and provides a phylogeny that contains sampled extinct tips and a table that contains the location of the edge samples. Let’s look at the setup of the edge samples table:

```
names(fbd.tree)
```

```
## [1] "phy"      "k.samples"
```

```
head(fbd.tree$k.samples)
```

```
##   taxon1 taxon2   timefrompresent
## 1    t18    t18 0.140477424088765
## 2   t128   t128 0.283143212511774
## 3   t147   t147 0.645288117495184
## 4    t12    t12 0.904722385034816
## 5     t3     t3 1.0610585239709
## 6    t85    t85 1.21026844132688
```

Internally, MiSSE will assume that the two taxa specified has an MRCA that is subtended by the branch where these edge fossils are located. The time from the present will know at what point on this branch this sample was taken. Note that if the branch from which a sample was taken is a tip branch, simply list the tip taxon twice (e.g., taxon1="sp12", taxon2="sp12"). Internally, MiSSE will place these fossils for you. For `hisse()` this table requires the state (if they are present):

```
k.samples.hisse <- data.frame(taxon1="sp12", taxon2="sp12", timefrompresent=3.164384,
                              state=1, stringsAsFactors=FALSE)
head(k.samples.hisse)
```

and `MuHiSSE()`,

```
k.samples.muhiisse <- data.frame(taxon1="sp12", taxon2="sp12", timefrompresent=3.164384,
                                 state1=0, state2=1, stringsAsFactors=FALSE)
head(k.samples.muhiisse)
```

Setting up a MiSSE model

Running `MiSSE()` (or `hisse()`, and `MuHiSSE()`) is essentially the same as before,

```
turnover <- c(1)
eps <- c(1)
one.rate <- MiSSE(fbd.tree$phy, f=1, turnover=turnover, eps=eps,
                  includes.fossils=TRUE, k.samples=fbd.tree$k.samples, sann=TRUE,
                  sann.its=1000)
```

Note that the only two additions to the function call are `includes.fossils=TRUE`, which indicates that the tree contains at least m fossils, and `k.samples=` that takes the table of edge samples. If no edge samples exist, this can be set to `NULL` and `hisse` will automatically default to the FBDT model:

```
one.rate <- MiSSE(fbd.tree$phy, f=1, turnover=turnover, eps=eps,
                  includes.fossils=TRUE, k.samples=NULL, sann=TRUE, sann.its=1000)
```

In such cases, the estimate of ψ will *only* reflect sampling of extinct tips. As we show in Beaulieu and O'Meara (2023), assuming both m and k fossils, but setting `k.samples=NULL` has a rather pronounced effect on the parameter estimates generally.

When reconstructing rates and/or states on the tree, the `MarginReconMiSSE()` also requires the same inputs:

```
margin.test <- MarginReconMiSSE(phy=fbd.tree$phy, f=1, pars=one.rate$solution,
                                hidden.states=1, includes.fossils=TRUE, k.samples=fbd.tree$k.samples,
                                aic=one.rate$AIC)
```

A cautionary tale about empirical trees and what is considered ‘surviving’

Trees containing fossil taxa are necessarily non-ultrametric. However, numerical differences in branch lengths can sometimes cause surviving tips to terminate slightly before the present, potentially resulting in their incorrect classification as extinct. We now provide a function called `CheckFossilTree()`, which allows users to check how tips will be classified by `hisse` and determine whether the default `extinct.tol` should be adjusted before fitting a fossil model. The `extinct.tol` argument is now also included in the function calls to `MiSSE()`, `hisse()`, and `MuHiSSE()`.

As an example, here is *Zamiaceae*, pruned from a much larger empirical tree of cycads from Coiro et al. (2023) that includes many fossil tips. Within this subtree, there should be 11 extinct tips and 111 surviving tips. However, if we were to run `MiSSE()` using the default tolerance of `extinct.tol=.Machine$double.eps^.50`, we get the following:

```
tree <- read.tree("zam.tre")
CheckFossilTree(tree, return.list=FALSE)
```

```
## Extant: 2
## Fossil: 120
```

Yikes – only 2 taxa are considered surviving here. If this check was never used and we ran `MiSSE()` we would probably notice that something is off because the estimate of ψ hits the upper bound of 1. However, since we did we can triage before fitting the model. Setting `return.list=TRUE` returns the estimated ages of the tips classified as surviving and extinct, which can help determine whether the default tolerance is too strict. In this case, many taxa known to be surviving have very small positive tip ages due to numerical differences in branch lengths. Here, `extinct.tol=.Machine$double.eps^.25` is a more appropriate tolerance:

```
tree <- read.tree("zam.tre")
CheckFossilTree(tree, extinct.tol=.Machine$double.eps^.25, return.list=FALSE)
```

```
## Extant: 111
## Fossil: 11
```

This returns the expected numbers of 111 surviving and 11 extinct tips. **Note** – if the tolerance is changed when calling `MiSSE()`, `hisse()`, or `MuHiSSE()`, the tolerance will also have to be adjusted reconstructing the states on the tree (e.g., `MarginReconMiSSE()`).

Using HisSE and MuHiSSE

The general structure for running `hisse()` and `MuHiSSE()` is identical to running `MiSSE()`. Below are examples that assume that the tree contains extinct taxa and a properly formatted `k.samples` table (assuming there are edge samples).

HiSSE:

```
trans.rate <- TransMatMakerHiSSE()
pp <- hisse(phy=fbd.tree$phy, data, f=c(1,1), turnover=c(1,1), eps=c(1,1),
            trans.rate=trans.rate, k.samples=fbd.tree$k.samples,
            includes.fossils=TRUE)
margin.test <- MarginReconHiSSE(phy=fbd.tree$phy, data=data, f=c(1,1),
                                pars=pp$solution, hidden.states=1, includes.fossils=TRUE,
                                k.samples=fbd.tree$k.samples)
```

MuHiSSE:

```
trans.rate <- TransMatMakerMuHiSSE()
pp <- MuHiSSE(phy=fbd.tree$phy, data, f=c(1,1,1,1), turnover=c(1,1,1,1),
              eps=c(1,1,1,1), trans.rate=trans.rate, includes.fossils=TRUE,
              k.samples=fbd.tree$k.samples)
margin.test <- MarginReconMuHiSSE(phy=fbd.tree$phy, data=data, f=c(1,1,1,1),
                                  pars=pp$solution, hidden.states=1, includes.fossils=TRUE,
                                  k.samples=fbd.tree$k.samples)
```

Stratigraphic ranges

We have also implemented the FBD model with stratigraphic ranges and I will show how to set up a standard run using the simulated data above. The first step is to take the fossil set sampled above and convert them to stratigraphic ranges. This can be done using a custom function within `hisse`:

```
strat.tree <- ProcessSimStrat(phy, f)
```

As with the `ProcessSimSample()`, `ProcessSimStrat()` takes the sampled fossil set and provides a phylogeny that contains sampled extinct tips and a table that contains the location of each stratigraphic range. The table of stratigraphic ranges requires a particular format:

```
names(strat.tree)
```

```
## [1] "phy"          "strat.intervals"
```

```
head(strat.tree$strat.intervals)
```

```
##   taxon1 taxon2 timefrompresentroot timefrompresenttip type
## 1   t18   t18   0.140477424088765 1.4210854715202e-14   R
## 2  t128  t128   1.93906488777661 0.283143212511774   R
## 3  t147  t147   0.645288117495184 7.105427357601e-15   R
## 4   t12   t12   0.904722385034816 1.4210854715202e-14   R
## 5    t3    t3   1.0610585239709 1.4210854715202e-14   R
## 6   t85   t85   1.21026844132688 1.4210854715202e-14   R
```

The columns `taxon1` and `taxon2` are two taxa specified whose MRCA is subtended by the branch where the stratigraphic range is located. The `timefrompresentroot` is the time at the rootward end stratigraphic range (denoted by o_i by Stadler et al. 2018) and `timefrompresenttip` is the time at the tipward end of range (denoted by y_i). Finally, the column `type` contains either an *R* for true stratigraphic ranges (i.e., $o_{\{i\}} - y_{\{i\}} > 0$) or an *S* for singleton fossils (i.e., $o_{\{i\}} - y_{\{i\}} = 0$). Note, as with the canonical FBD

implementation above, if the branch from which a range was sampled is a tip branch, simply list the tip taxon twice (e.g., taxon1="sp12", taxon2="sp12").

A run proceeds exactly as before, except instead we supply the stratigraphic range table to the "strat.interval" option in the function call:

```
turnover <- c(1)
eps <- c(1)
one.rate <- MiSSE(strat.tree$phy, f=1, turnover=turnover, eps=eps,
  includes.fossils=TRUE, k.samples=NULL,
  strat.intervals=strat.tree$strat.intervals, sann=TRUE,
  sann.its=5000)
```

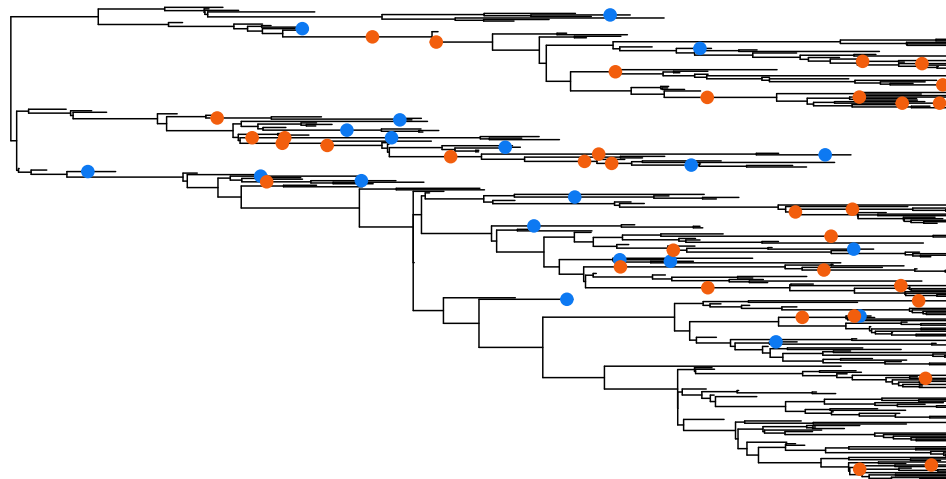
When reconstructing rates and/or states on the tree, the `MarginReconMiSSE()` also requires the same inputs:

```
margin.test <- MarginReconMiSSE(phy=strat.tree$phy, f=1, pars=one.rate$solution,
  hidden.states=1, includes.fossils=TRUE, k.samples=NULL,
  strat.intervals=strat.tree$strat.intervals, aic=one.rate$AIC)
```

Plotting simulated data sets

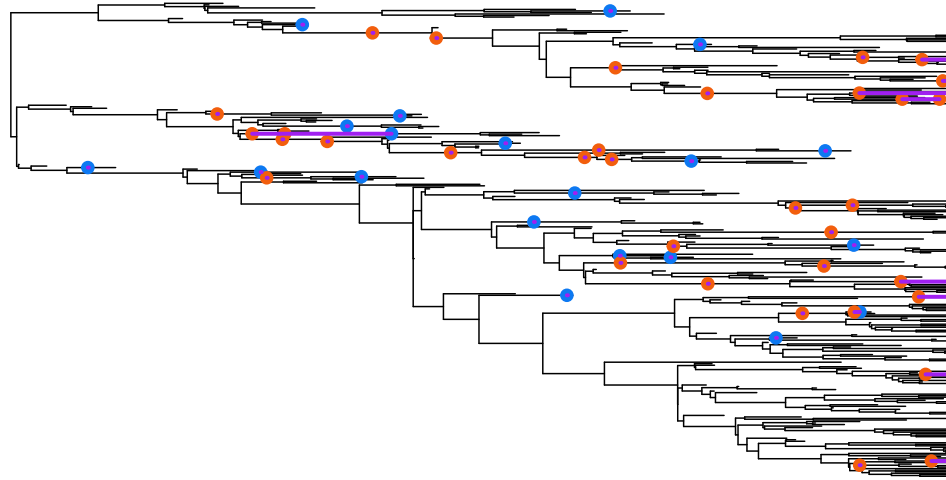
Plotting functions of the reconstructions work as they did before and do not require any additional arguments. However, we have added functionality for overlaying the fossil set and stratigraphic ranges obtained from our simulation functions. For example, here I will plot the full simulated tree from above, and then add the m and k fossils as points onto of the phylogeny:

```
plot(ladderize(phy), show.tip.label=FALSE, edge.width=0.75)
### Split the table into m and k for the points ###
extinct.samples <- f[which(f$fossiltype_long=="extinct_terminal" |
  f$fossiltype_long=="extinct_internal"),]
k.samples.tmp <- extinct.samples[which(extinct.samples$has_sampled_descendant == TRUE),]
extinct.samples <- extinct.samples[which(extinct.samples$has_sampled_descendant == FALSE),]
k.samples <- f[which(f$fossiltype_long == "surviving_terminal" |
  f$fossiltype_long == "surviving_internal"),]
k.samples <- rbind(k.samples, k.samples.tmp)
AddFossilPoints(ladderize(phy), f=extinct.samples, pch=19, cex=0.8,
  col="#0D79F2")
AddFossilPoints(ladderize(phy), f=k.samples, pch=19, cex=0.8, col="#F25E0D")
```



We can also overlay the lines designating stratigraphic ranges:

```
plot(ladderize(phy), show.tip.label=FALSE, edge.width=0.75)
AddFossilPoints(ladderize(phy), f=extinct.samples, pch=19, cex=0.8,
               col="#0D79F2")
AddFossilPoints(ladderize(phy), f=k.samples, pch=19, cex=0.8, col="#F25E0D")
AddStratIntervals(ladderize(phy), f=f, pch=19, cex=0.8, col="purple", lwd=2)
```



Checking the MiSSE, HiSSE, and MuHiSSE likelihoods

The easiest way to check that the likelihoods calculated by `MiSSE()`, `hisse()`, and `MuHiSSE()` for the standard FBD are correct is to take the likelihood calculated by `MiSSE()` and add this to the likelihood of a simple Markov transition model. As described in Caetano et al. (2018), an SSE model jointly maximizes the probability of the observed states at the tips *and* the observed tree, given the model. Thus, if the character is completely disassociated from the rate differences in the tree (with what we call a character independent model) then this test should work.

I will show this test here. First, I will calculate the likelihood of a simple BiSSE model, where the diversification rates are the same for states 0 and 1, on a simulated dataset that contains fossils:

```
library(diversitree)
pars <- c(0.1, 0.2, 0.03, 0.03, 0.01, 0.01)
set.seed(4)
phy <- NULL
while( is.null( phy ) ){
  phy <- tree.bisse(pars, max.t=30, x0=0, include.extinct=TRUE)
}
k.samples <- data.frame(taxon1="sp12", taxon2="sp12", timefrompresent=3.164384,
                        state=1, stringsAsFactors=FALSE)
hidden.states=FALSE
phy.k <- hisse:::AddKNodes(phy, k.samples)
fix.type <- hisse:::GetKSampleMRCA(phy.k, k.samples)
nb.tip <- Ntip(phy.k)
```

```

nb.node <- phy.k$Nnode
gen <- hisse::FindGenerations(phy.k)

data <- data.frame(taxon=names(phy$tip.state), phy$tip.state,
  stringsAsFactors=FALSE)
data <- hisse::AddKData(data, k.samples)
data.new <- data.frame(data[,2], data[,2], row.names=data[,1])
data.new <- data.new[phy.k$tip.label,]

dat.tab <- hisse::OrganizeDataHiSSE(data.new, phy=phy.k, f=c(1,1),
  hidden.states=FALSE, includes.fossils=TRUE)
edge_details <- hisse::GetEdgeDetails(phy=phy.k,
  intervening.intervals=strat.cache$intervening.intervals)
fossil.taxa <- edge_details$tipward_node[which(edge_details$type == "extinct_tip")]
pars.bisse <- c(0.1+0.03, 0.1+0.03, 0.03/0.1, 0.03/0.1, 0.01, 0.01)

model.vec <- numeric(48)
model.vec[1:6] = pars.bisse
phy$node.label = NULL
cache <- hisse::ParametersToPassHiSSE(model.vec, hidden.states=hidden.states,
  nb.tip=Ntip(phy.k), nb.node=Nnode(phy.k), psi.type="m+k",
  bad.likelihood=-300, f=c(1,1), ode.eps=0)
cache$psi <- 0.01
hisse.full <- hisse::DownPassHiSSE(dat.tab, gen, cache, root.type="madfitz",
  condition.on.survival=TRUE, root.p=NULL, node=fix.type$node,
  state=fix.type$state, fossil.taxa=fossil.taxa,
  fix.type=fix.type$type)

```

Next we will calculate the likelihood of *just* the tree using MiSSE():

```

dat.tab <- hisse::OrganizeDataMiSSE(phy=phy.k, f=1, hidden.states=1, includes.fossils=TRUE)
model.vec <- c(0.1+0.03, 0.03/0.1, rep(0,51))
cache = hisse::ParametersToPassMiSSE(model.vec=model.vec, hidden.states=1,
  fixed.eps=NULL, nb.tip=nb.tip, nb.node=nb.node,
  psi.type="m+k", bad.likelihood=exp(-500), ode.eps=0)#
cache$psi <- 0.01
gen <- hisse::FindGenerations(phy.k)
MiSSE.logL <- hisse::DownPassMisse(dat.tab=dat.tab, cache=cache, gen=gen,
  condition.on.survival=TRUE, root.type="madfitz", root.p=NULL,
  fossil.taxa=fossil.taxa, node=fix.type$node, fix.type=fix.type$type)

```

Finally, I will use corHMM to calculate the likelihood of *just* character data:

```

library(corHMM)
char.logL <- corHMM(phy.k, data, rate.cat=1, model = "ER", node.states = "none",
  fixed.nodes=FALSE, p=0.01, root.p="maddfitz")

```

You specified 'fixed.nodes=FALSE' but included a phy object with node labels. These node labels have

```

## Warning in corHMM(phy.k, data, rate.cat = 1, model = "ER", node.states =
## "none", : Branch lengths of 0 detected. Adding 1e-5 to these branches.

```

```
## State distribution in data:
## States: 1 2
## Counts: 2 17
## Calculating likelihood from a set of fixed parameters
```

We can compare the likelihoods obtained from `hisse()` against the sum of the tree *and* the character:

```
tot.logL <- char.logL$loglik + MiSSE.logL
comparison <- identical(round(hisse.full,3), round(tot.logL,3))
comparison
```

```
## [1] TRUE
```

This confirms that the calculations are correct. I will show the same using `MuHisSE()`:

```
library(diversitree)
pars <- c(.1, .15, .2, .1,
          .03, .045, .06, 0.03,
          .05, .05, .00,
          .05, .00, .05,
          .05, .00, .05,
          .00, .05, .05)
set.seed(2)
phy <- NULL
while( is.null( phy ) ){
  phy <- tree.musse(pars, 30, x0=1, include.extinct=TRUE)
}
k.samples <- data.frame(taxon1="sp20", taxon2="sp37", timefrompresent=8.54554,
                        state1=0, state2=1, stringsAsFactors=FALSE)

phy.k <- hisse::AddKNodes(phy, k.samples)
fix.type <- hisse::GetKSampleMRCA(phy.k, k.samples)
nb.tip <- Ntip(phy.k)
nb.node <- phy.k$Nnode
gen <- hisse::FindGenerations(phy.k)

states <- phy$tip.state
states <- data.frame(phy$tip.state, phy$tip.state,
row.names=names(phy$tip.state))
states <- states[phy$tip.label,]
states.trans <- states
for(i in 1:Ntip(phy)){
  if(states[i,1] == 1){
    states.trans[i,1] = 0
    states.trans[i,2] = 0
  }
  if(states[i,1] == 2){
    states.trans[i,1] = 0
    states.trans[i,2] = 1
  }
  if(states[i,1] == 3){
    states.trans[i,1] = 1
    states.trans[i,2] = 0
  }
}
```

```

    }
    if(states[i,1] == 4){
      states.trans[i,1] = 1
      states.trans[i,2] = 1
    }
  }

data <- data.frame(taxon=names(phy$tip.state),
                  states.trans[,1], states.trans[,2], stringsAsFactors=FALSE)
data <- hisse:::AddKData(data, k.samples, muhisse=TRUE)
data.new <- data.frame(data[,2], data[,3], row.names=data[,1])
data.new <- data.new[phy.k$tip.label,]

pars.muhisse <- c(rep(0.1+0.03,4), rep(0.03/.1, 4), 0.05,0.05,0, 0.05,0,0.05,
                  0.05,0,.05, 0,0.05,.05)
model.vec = rep(0,384)
model.vec[1:20] = pars.muhisse
cache <- hisse:::ParametersToPassMuHiSSE(model.vec=model.vec, hidden.states=FALSE,
    psi.type="m+k", nb.tip=Ntip(phy.k), nb.node=Nnode(phy.k),
    bad.likelihood=exp(-500), f=c(1,1,1,1), ode.eps=0)
cache$psi <- 0.01
gen <- hisse:::FindGenerations(phy.k)
dat.tab <- hisse:::OrganizeData(data.new, phy.k, f=c(1,1,1,1),
    hidden.states=FALSE, includes.fossils=TRUE)
fossil.taxa <- which(dat.tab$branch.type == 1)

muhisse.full <- hisse:::DownPassMuHisse(dat.tab, gen=gen, cache=cache,
    root.type="madfitz", condition.on.survival=TRUE, root.p=NULL,
    node=fix.type$node, state=fix.type$state, fossil.taxa=fossil.taxa,
    fix.type=fix.type$type)

## Trait independent model should be loglik_tree + loglik_character ##
#Part 1: MiSSE loglik:
dat.tab <- hisse:::OrganizeDataMiSSE(phy=phy.k, f=1, hidden.states=1, includes.fossils=TRUE)
model.vec <- c(0.1+0.03, 0.03/0.1, rep(0,51))
cache = hisse:::ParametersToPassMiSSE(model.vec=model.vec, hidden.states=1,
    fixed.eps=NULL, nb.tip=nb.tip, nb.node=nb.node,
    psi.type="m+k", bad.likelihood=exp(-500), ode.eps=0)#
cache$psi <- 0.01
edge_details <- hisse:::GetEdgeDetails(phy=phy.k,
    intervening.intervals=strat.cache$intervening.intervals)
fossil.taxa <- edge_details$tipward_node[which(edge_details$type == "extinct_tip")]
gen <- hisse:::FindGenerations(phy.k)
MiSSE.logL <- hisse:::DownPassMisse(dat.tab=dat.tab, cache=cache, gen=gen,
    condition.on.survival=TRUE, root.type="madfitz", root.p=NULL,
    fossil.taxa=fossil.taxa,node=fix.type$node, fix.type=fix.type$type)

#Part 2: corHMM loglik:
phy.k$node.label <- NULL
char.logL <- corHMM(phy.k, data, rate.cat=1, model = "ER", node.states = "none",
    fixed.nodes=FALSE, p=0.05, root.p="maddfitz")

```

```
## Warning in corHMM(phy.k, data, rate.cat = 1, model = "ER", node.states =
```

```
## "none", : Branch lengths of 0 detected. Adding 1e-5 to these branches.
```

```
## State distribution in data:
```

```
## States: 1 2 3 4
```

```
## Counts: 12 21 6 3
```

```
## Calculating likelihood from a set of fixed parameters
```

```
tot.logL <- char.logL$loglik + MiSSE.logL
```

```
comparison <- identical(round(muhisse.full,3), round(tot.logL,3))
```

```
comparison
```

```
## [1] TRUE
```

References

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Coiro, M., R. Allio, N. Mazet, L.J. Seyfullah, and F.L. Condamine. (2023). Reconciling fossils with phylogenies reveals the origin and macroevolutionary processes explaining the global cycad biodiversity. *New Phytology*, 240:1616-1635.

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